

# Classifying quadrilaterals



- 1
- a Rectangle
  - b Square
  - c Trapezoid
  - d Rectangle
  - e Trapezoid
- 2
- a Rhombus
  - b Parallelogram
  - c Trapezoid
  - d Rhombus
  - e Trapezoid
  - f Parallelogram
- 3 Parallelogram
- 4
- a False
  - b True
  - c False
  - d False
  - e False
  - f True
  - g False
  - h True
  - i False
  - j True
  - k True
  - l False
  - m False
- 5
- a Rectangle
  - b Parallelogram
  - c Rhombus
  - d Rectangle
  - e Rhombus
  - f Parallelogram
  - g Rhombus
  - h Parallelogram
- 6
- a Parallelogram
  - b Rectangle
- 7
- a There is an additional pair of parallel sides.
  - b Two pairs of opposite sides are congruent.
  - c One pair of opposite sides are parallel.
- 8
- a The parallelogram has four congruent sides.
  - b The parallelogram has four right angles.
  - c The parallelogram has four congruent sides and four right angles.

9



- a The trapezoid has diagonals that are congruent.
- b The trapezoid has base angles that are congruent.
- c The trapezoid has opposite angles that are supplementary.

10

**To prove:  $ABCD$  is a square**

| Statements                                             | Reasons                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1. $ABCD$ is a rectangle and a rhombus                 | Given                                                                                                 |
| 2. $ABCD$ is a parallelogram with four right angles    | Definition of a rectangle                                                                             |
| 3. $ABCD$ is a parallelogram with four congruent sides | Definition of a rhombus                                                                               |
| 4. $ABCD$ is a square                                  | A parallelogram with four congruent sides and four right angles is a square (definition of a square). |

11

**To prove:  $ABCD$  is a rectangle**

| Statements                                                                                         | Reasons                                                                                                                                   |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1. $ABCD$ is a parallelogram and $\overline{AC} \cong \overline{BD}$                               | Given                                                                                                                                     |
| 2. $\overline{AD} \cong \overline{BC}$                                                             | If a quadrilateral is a parallelogram, then it has both pairs of opposite sides congruent.                                                |
| 3. $\overline{AB} \cong \overline{AB}$ and $\overline{CD} \cong \overline{CD}$                     | Reflexive property of congruence                                                                                                          |
| 4. $\triangle ABD \cong \triangle BAC$ and $\triangle ACD \cong \triangle BDC$                     | Side-side-side congruence theorem                                                                                                         |
| 5. $\angle ADC \cong \angle BCD$ and $\angle DAB \cong \angle CBA$                                 | Corresponding parts of congruent triangles are congruent (CPCTC).                                                                         |
| 6. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$             | Definition of a parallelogram                                                                                                             |
| 7. $\angle ADC$ is supplementary to $\angle BCD$ and $\angle DAB$ is supplementary to $\angle CBA$ | If two parallel lines are cut by a transversal, then consecutive interior angles are supplementary (consecutive interior angles theorem). |
| $m\angle ADC + m\angle BCD = 180^\circ$ and                                                        |                                                                                                                                           |



| Statements |                                                                                        | Reasons                       |
|------------|----------------------------------------------------------------------------------------|-------------------------------|
| 9.         | $m\angle ADC + m\angle ADC = 180^\circ$ and<br>$m\angle DAB + m\angle DAB = 180^\circ$ | Substitution                  |
| 10.        | $2m\angle ADC = 180^\circ$ and<br>$2m\angle DAB = 180^\circ$                           | Simplify                      |
| 11.        | $m\angle ADC = 90^\circ$ and<br>$m\angle DAB = 90^\circ$                               | Division property of equality |
| 12.        | $\angle ADC, \angle BCD, \angle DAB$ and $\angle CBA$ are<br>right angles              | Definition of a right angle   |
| 13.        | $ABCD$ is a rectangle                                                                  | Definition of a rectangle     |

12  
a

| To prove: $ABCD$ is a rhombus |                                                                             |                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Statements                    |                                                                             | Reasons                                                                          |
| 1.                            | $ABCD$ is a parallelogram and<br>$\overline{AB} \cong \overline{BC}$        | Given                                                                            |
| 2.                            | $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$ | If a quadrilateral is a parallelogram,<br>then its opposite sides are congruent. |
| 3.                            | $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$ | Transitive property of congruence                                                |
| 4.                            | $ABCD$ is a rhombus                                                         | Definition of a rhombus                                                          |

b

To prove:  is a rhombus

| Statements                                                                     | Reasons                                                                                   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1. $PQRS$ is a parallelogram and $\overline{PR} \perp \overline{QS}$           | Given                                                                                     |
| 2. $\overline{PR}$ and $\overline{QS}$ bisect each other                       | If a quadrilateral is a parallelogram, then it has two diagonals that bisect one another. |
| 3. $\overline{PT} \cong \overline{TR}$ and $\overline{QT} \cong \overline{TS}$ | Definition of a bisector                                                                  |
| 4. $\angle STP, \angle QTP, \angle QTR, \angle RTS$ are right angles           | Definition of perpendicular                                                               |
| 5. $\angle STP \cong \angle QTP \cong \angle QTR \cong \angle RTS$             | Definition of congruent angles                                                            |
| 6. $\triangle STP \cong \triangle QTP \cong \triangle QTR \cong \triangle RTS$ | Side-angle-side congruence theorem                                                        |
| 7. $\overline{PQ} \cong \overline{QR} \cong \overline{RS} \cong \overline{ST}$ | Corresponding parts of congruent triangles are congruent (CPCTC).                         |
| 8. $PQRS$ is a rhombus                                                         | Definition of a rhombus                                                                   |

c


 To prove:  $ABCD$  is a rhombus

| Statements                                                                             | Reasons                                                                                                               |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1. $ABCD$ is a parallelogram and $\overline{BD}$ bisects $\angle ABC$ and $\angle ADC$ | Given                                                                                                                 |
| 2. $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$                             | Definition of an angle bisector                                                                                       |
| 3. $\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{CB}$ | Definition of a parallelogram                                                                                         |
| 4. $\angle 1 \cong \angle 3$ and $\angle 2 \cong \angle 4$                             | If two lines are parallel, then alternate interior angles are congruent (alternate interior angles theorem).          |
| 5. $\angle 1 \cong \angle 4$ and $\angle 2 \cong \angle 3$                             | Transitive property of congruence                                                                                     |
| 6. $\overline{DA} \cong \overline{AB}$ and $\overline{BC} \cong \overline{CD}$         | If two angles of a triangle are congruent, then the sides opposite those angles are congruent (base angles converse). |
| 7. $\overline{AB} \cong \overline{DC}$ and $\overline{AD} \cong \overline{BC}$         | If a quadrilateral is a parallelogram, then the opposite sides are congruent.                                         |
| 8. $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$         | Transitive property of congruence                                                                                     |
| 9. $ABCD$ is a rhombus                                                                 | Definition of a rhombus                                                                                               |

13

a

To prove:  $ABCD$  is an isosceles trapezoid

| Statements                                                                                                      | Reasons                                                          |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1. $ABCD$ is a trapezoid,<br>$\angle ADC \cong \angle BCD$ , and $\angle APD$ and $\angle BQC$ are right angles | Given                                                            |
| 2. $\angle APD \cong \angle BQC$                                                                                | All right angles are congruent.                                  |
| 3. $\overline{AP} \cong \overline{BQ}$                                                                          | Altitudes of a trapezoid are congruent                           |
| 4. $\triangle APD \cong \triangle BQC$                                                                          | Angle-angle-side congruence theorem                              |
| 5. $\overline{AD} \cong \overline{BC}$                                                                          | Corresponding parts of congruent triangles are congruent (CPCTC) |
| 6. $ABCD$ is an isosceles trapezoid                                                                             | Definition of an isosceles trapezoid                             |

b

To prove:  $ABCD$  isosceles trapezoid

| Statements                                                            | Reasons                                                                                                                |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 1. $ABCD$ is a trapezoid, $\angle A$ and $\angle C$ are supplementary | Given                                                                                                                  |
| 2. $\overline{AB} \parallel \overline{DC}$                            | Definition of a trapezoid                                                                                              |
| 3. $\angle A$ and $\angle D$ are supplementary                        | If two lines are parallel, then consecutive interior angles are supplementary (consecutive interior angles postulate). |
| 4. $\angle C \cong \angle D$                                          | Supplements of the same angle or congruent angles are congruent (congruent supplements theorem).                       |
| 5. $ABCD$ is an isosceles trapezoid                                   | If a trapezoid has base angles congruent, then it is an isosceles trapezoid.                                           |

c

To prove:  $ABCD$  isosceles trapezoid

| Statements                                                                                                | Reasons                                                                                                             |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1. $ABCD$ is a trapezoid, $\overline{AC} \cong \overline{BD}$ and $\overline{AC} \parallel \overline{BP}$ | Given                                                                                                               |
| 2. $\overline{AB} \parallel \overline{DC}$                                                                | Definition of a trapezoid                                                                                           |
| 3. $ABPC$ is a parallelogram                                                                              | Definition of a parallelogram                                                                                       |
| 4. $\overline{AC} \cong \overline{BP}$                                                                    | If a quadrilateral is a parallelogram, then its opposite sides are congruent.                                       |
| 5. $\overline{BD} \cong \overline{BP}$                                                                    | Transitive property of congruence                                                                                   |
| 6. $\angle BDC \cong \angle BPC$                                                                          | If two sides of a triangle are congruent, then the angles opposite those sides are congruent (base angles theorem). |
| 7. $\angle ACD \cong \angle BPC$                                                                          | If two lines are parallel, then corresponding angles are congruent (corresponding angles theorem).                  |
| 8. $\angle BDC \cong \angle ACD$                                                                          | Transitive property of congruence                                                                                   |
| 9. $\overline{DC} \cong \overline{DC}$                                                                    | Reflexive property of congruence                                                                                    |
| 10. $\triangle ACD \cong \triangle BDC$                                                                   | Side-angle-side congruence theorem (SAS)                                                                            |
| 11. $\overline{AD} \cong \overline{BC}$                                                                   | Corresponding parts of congruent triangles are congruent (CPCTC)                                                    |
| 12. $ABCD$ is an isosceles trapezoid                                                                      | Definition of an isosceles trapezoid                                                                                |

## Additional questions

14

- |                                |                                           |
|--------------------------------|-------------------------------------------|
| a Kite                         | b Rectangle, rhombus, parallelogram       |
| c Rhombus, kite, parallelogram | d Parallelogram                           |
| e Kite                         | f Rectangle, rhombus, kite, parallelogram |
| g Kite                         | h Rectangle, rhombus, parallelogram       |
| i Parallelogram                | j Trapezoid                               |
| k Neither                      | l Parallelogram                           |



15 3, because otherwise the angle sum would be larger than 360.

16 4, this is the maximum number of angles a quadrilateral can have, and it would give the correct angle sum. Squares and rectangles have four right-angles.

17 Yes, a trapezium.

18 No